## Preparation list for 2017 JC2 H2 Biology Prelim Paper 4 -- Practical 2017

### Question 1

| Apparatus/Reagents/Chemicals | Quantity per student |
|---|---|
| Potato extract, labelled **T**, at room temperature | 30 $cm^3$ |
| 1.0% amylase solution in a container, labelled **E**, at room temperature | 10 $cm^3$ |
| 1.0% starch solution in a container, labelled **S**, at room temperature | 30 $cm^3$ |
| 1 $mol dm^{-3}$ sulfuric acid in a container, labelled **A**, at room temperature | 20 $cm^3$ |
| 0.1% potassium manganate (VII) solution in a container, labelled **P**, at room temperature | 20 $cm^3$ |
| Iodine solution in container with a dropping pipette or means to remove it, labelled **iodine**, at room temperature | 15 $cm^3$ |
| 10 $cm^3$ syringe with the means to wash them out | 1 |
| 3 $cm^3$ or 5 $cm^3$ syringes with the means to wash them out | 2 |
| Spatula | 1 |
| Dropping pipettes | 2 |
| Glass microscope slides | 1 |
| Coverslips | 2 |
| Sieve | 1 |
| Beaker or container, capacity sufficient to hold approximately 100 $cm^3$ solution | 2 |
| Test tubes, small | 6 |
| Test tube rack to hold 6 test tubes | 1 |
| Glass rod | 1 |
| Mounting needle | 1 |
| Spotting tile or white tile | 1 |
| 15 cm ruler | 1 |
| Paper towels | 5 |
| Glass marker pen | 1 |
| Stopwatch | 1 |
| Suitable eye protection | 1 |

**Question 1**

It is advisable to wear suitable eye protection when handling chemicals.

**Preparation of solutions**

(i) **T**, potato extract
This is prepared by juicing well washed potatoes.
**T** must be prepared immediately before the examination.

(ii) **E**, 1.0% amylase solution
This prepared by putting 1 g of amylase powder into a beaker and making up to 100 $cm^3$ with distilled water and mixing well.
**E** must be prepared immediately before the examination.

(iii) **S**, 1.0% starch solution
This is prepared by putting 1 g of starch into 25 $cm^3$ of warm distilled water in a beaker and mixing to a paste, making up to 100 $cm^3$ with boiling distilled water, Mix well and allow to cool.

(iv) **A**, 1.0 $moldm^{-3}$ sulfuric acid
This is prepared from (98%) sulfuric acid, by adding 5 $cm^3$ of this sulfuric acid to 500 $cm^3$ of distilled water and making up to 1 $dm^3$ with distilled water.
This is an exothermic reaction, **add acid to water**.

(v) **P**, 0.1% potassium manganate (VII) solution
This is prepared by putting 1.0 g of potassium manganate (VII) into a beaker and making up to 100 $cm^3$ with distilled water. This is to make a 1.0% solution.
Then put 10 $cm^3$ of this 1.0% solution into a beaker and make up to 100 $cm^3$ with distilled water.
This solution must be made up immediately before the start of the examination and kept out of sunlight.

(vi) **iodine**, iodine solution (0.1 $moldm^{-3}$)
This is prepared by putting 8 g of potassium iodide into a beaker. Moisten the potassium iodide with a few drops of water. Add 2.54 g of iodine to the potassium iodide and stir well. Make up to 100 $cm^3$ adding small volumes of distilled water and stir well. Continue to stir until the iodine has dissolved.
This solution must be made up immediately before the start of the examination and kept out of sunlight.

**Question 2**

Each candidate must have sole, uninterrupted use of a microscope for 1 hour 15 minutes only.

| Apparatus | Quantity per student |
|---|---|
| Stage micrometer with divisions down to 0.1mm or 0.01mm | at least 1 between 2 |
| Prepared slide of Leaf of Rosemary (TS), labelled **S1** | at least 1 between 2 |